

Smart Contract Audit Report

AlphaAdvisor (APAD) BEP-20 Token, BNB Smart Chain

A review of the deployed StandardBEP20 contract behind the APAD token: verified source against deployed bytecode, the callable surface, the owner role and the concentration of the supply. The audit was performed and published in December 2025. This edition keeps the original findings and re-verifies every factual statement against the live chain on 22 September 2026, nine months after the original report.

OVERALL RISK RATING **LOW**

PREPARED BY SOKEN LTD, Smart Contract Security	ORIGINAL REPORT December 2025, published 8 December 2025
THIS EDITION 22 September 2026, figures re-verified	STATE READ AT block 123,364,658
NETWORK BNB Smart Chain (chainId 56)	FINDINGS 0 Critical, 0 High, 0 Medium, 2 Low, 2 Informational
TOKEN CONTRACT 0x51b27860c3432a72e24a7e0a98916d10cb716d5c	COMPILER solc 0.8.30, optimizer disabled, EVM prague

SOKEN LTD. Security review of on-chain source code. Not financial, investment or legal advice. See the disclaimer in section 09.

01 Executive summary

APAD is a fixed-supply BEP-20 token deployed on 29 August 2025 from a standard token template. The project-specific contract, StandardBEP20, combines the unmodified OpenZeppelin 5.1.0 ERC20 base with an immutable decimals value and an OpenZeppelin Ownable owner exposed through the BEP-20 `getOwner()` method. The whole supply of 1,000,000,000 APAD is minted once, in the constructor, to the deploying account. There is no mint or burn path after deployment, no fee, no pause, no blacklist and no upgrade mechanism. We re-confirmed each absence by calling the selectors against the live contract.

The audit found no defect in the token logic. The two Low findings are about control rather than code: the owner role sits with a single externally owned account, and the full supply was minted to one account with no on-chain vesting. The Informational items are the standard ERC-20 approval race and, added in this edition, a payable constructor with no way to withdraw native coin.

- **The deployed code is the verified code.** Sourcify reports a full creation and runtime match for this address. Our own recompilation with solc 0.8.30 reproduced the same 4,277-byte runtime; the executable code is identical once the decimals immutable is filled in.
- **Supply is fixed.** `totalSupply()` returns 1,000,000,000 APAD today, the amount minted at deployment.
- **Control has changed since the audit, but not its shape.** On 13 February 2026 the deployer moved the full supply out, and on 14 February 2026 it transferred ownership to another single externally owned account, which today also holds 795,929,872 APAD (79.59% of supply). The owner still cannot touch any holder's balance.
- **Re-verification suite.** 41 Foundry cases (36 unit cases and 5 fuzz properties at 512 runs each) were written against the verified source and executed on 22 September 2026: 41 of 41 passing. Slither reported no High or Medium item.

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CRITICAL, HIGH AND MEDIUM FINDINGS

2 + 2

LOW AND INFORMATIONAL FINDINGS

1e27

BASE UNITS, SUPPLY UNCHANGED

41 / 41

RE-VERIFICATION TESTS PASSING

full match

SOURCIFY, CREATION AND RUNTIME

79.59%

OF SUPPLY IN THE CURRENT OWNER ACCOUNT

02 Scope, method and re-verification

AREA	WHAT WE DID
Source under audit	One flattened file, StandardBEP20.sol, 856 lines, as verified for this address. It contains OpenZeppelin Context 5.0.1, Ownable 5.0.0, ERC20, IERC20, IERC20Metadata and IERC6093 at 5.1.0, and the template contracts BEP20TokenOwner, ERC20Decimals, ERC20Detailed and StandardBEP20. The file also declares IPayable and ServicePayer, which the deployed contract does not inherit or use.
Deployed code identity	Sourcify: creationMatch = match, runtimeMatch = match (verified 18 March 2026). Local recompilation from the same standard-JSON input with solc 0.8.30+commit.73712a01, optimizer disabled, EVM target prague: identical executable code; the source hash matches the on-chain metadata.
Live surface probe	Called mint, burn, burnFrom, pause, paused, blacklist, upgradeTo, implementation, DOMAIN_SEPARATOR, nonces and increaseAllowance against the live contract. All revert. Read owner, getOwner, balances and supply at block 123,364,658.

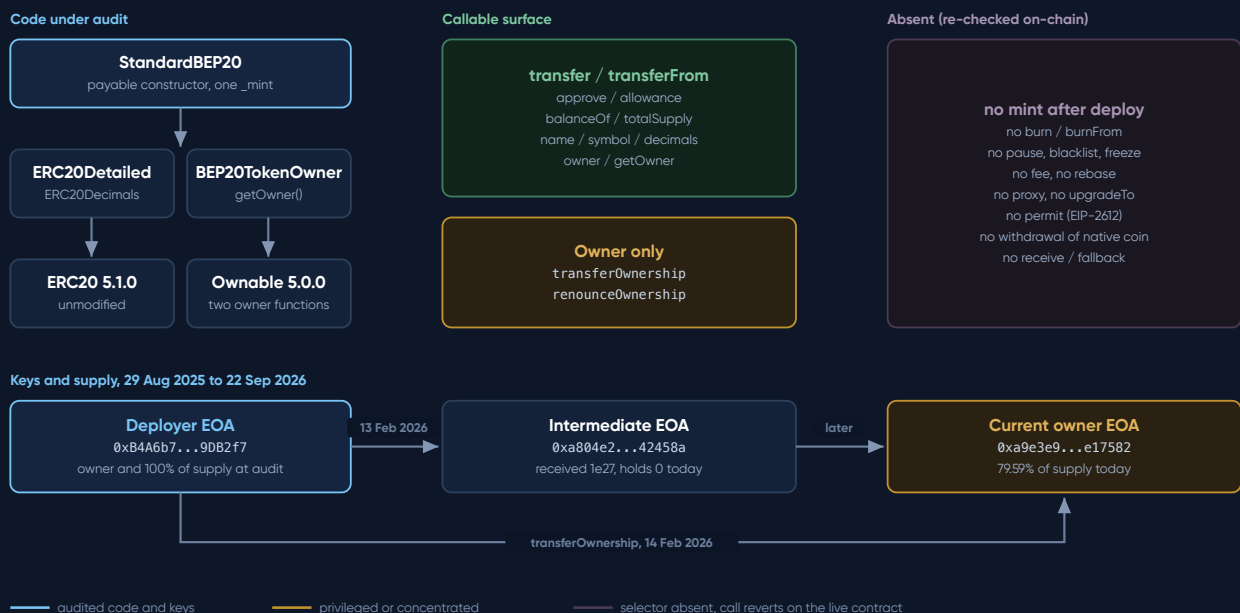
History	Located the ownership transfer and the first movement of the supply by binary search over archive state, then read the transactions in the blocks concerned.
Behavioural testing	41 Foundry cases: metadata and supply, transfer and allowance semantics, ERC-6093 errors, owner restrictions, absence of administrative selectors, the payable constructor, recipient callbacks, and 5 fuzz properties over supply, balances, allowances and ownership.
Static analysis	Slither 0.11.5: 5 results (1 locked-ether, 3 dead-code, 1 solc-version), no High or Medium.

Out of scope

Key management of the owner and holder accounts, exchange listings and liquidity, the AlphaAdvisor application, and the project's tokenomics and vesting documents. The contract enforces no vesting, so any schedule depends on off-chain conduct that this review does not cover.

03 Contract surface and control

What the contract exposes, and where control and supply sit



Top: inheritance, callable surface and absent functions. Bottom: the two accounts that have held the owner role and the supply, with the dates read from archive state. Addresses are shortened here and given in full in section 06.

ACTOR	POWER ON-CHAIN
Owner 0xA9e3e9...e17582	May call <code>transferOwnership</code> and <code>renounceOwnership</code> , nothing else. The role is visible to BEP-20 tooling through <code>getOwner()</code> . It gives no power over balances, supply, transfers or metadata.
Token holders	Transfer their own balance, grant and revoke allowances.
Deployer 0xB4A6b7...9DB2f7	No remaining role and a zero APAD balance. The account now carries an EIP-7702 delegation; this does not affect the token.

04 Findings

ID	FINDING	SEVERITY	STATUS
SKN-022	Owner role held by a single externally owned account	■ LOW	■ OPEN
SKN-143	Entire supply minted to one account, no on-chain vesting	■ LOW	■ OPEN
SKN-230	Standard ERC-20 approval race	■ INFORMATIONAL	■ ACKNOWLEDGED
A-04	Payable constructor with no withdrawal path (added in this edition)	■ INFORMATIONAL	■ ACKNOWLEDGED

SKN-022 Owner role held by a single externally owned account

■ LOW

Location. StandardBEP20.sol:747, the BEP20TokenOwner constructor, which sets `ownable(_msgSender())`.

At the audit date the owner was the deploying account `0xB4A6b7...9DB2f7`, a plain key. The owner can only transfer or renounce the role, so a compromised key cannot seize balances, mint or freeze. The practical risk is reputational and procedural: BEP-20 tooling reports `getOwner()` as the accountable party, and a stolen key would let an attacker present themselves as that party.

Current position. Ownership was transferred on 14 February 2026 08:53 UTC in `0x87938001...899f07f8` to `0xa9e3e9...e17582`, which has no contract code and is therefore also a single key. The recommendation below still applies.

Recommendation. Move the role to a multi-signature wallet, or renounce it if no owner function is needed.

Status. Open.

SKN-143 Entire supply minted to one account, no on-chain vesting

■ LOW

Location. StandardBEP20.sol:854, `_mint(_msgSender(), initialBalance_)` in the constructor.

All 1,000,000,000 APAD were minted to the deployer in `0xaa8c3009...37f7933d` at block 59,264,389, 29 August 2025 05:49 UTC. Archive state shows the deployer held the full supply without interruption until block 80,907,460, so it held 100% on the audit date. The contract has no vesting or lock logic, so any allocation schedule is enforced off-chain.

Current position. On 13 February 2026 03:40 UTC the deployer sent the full supply to `0xa804e2...42458a` in `0x5fb72709...0ca88895`. That account holds nothing today. At block 123,364,658 the current owner account holds 795,929,872 APAD (79.59% of supply). We have not traced every intermediate transfer and make no statement about the purpose of these movements.

Recommendation. Hold unreleased allocations in audited vesting or timelock contracts and publish the address that holds each allocation, so the distribution can be checked on-chain.

Status. Open.

SKN-230 Standard ERC-20 approval race

■ INFORMATIONAL

Location. Inherited approve from OpenZeppelin ERC20 5.1.0.

Changing a non-zero allowance to another non-zero value lets a spender who watches the mempool use both the old and the new allowance. OpenZeppelin 5.x removed `increaseAllowance` and `decreaseAllowance`, and the live contract rejects both selectors.

Recommendation. Wallets and integrators should reset an allowance to zero before setting a new value.

Status. Acknowledged. Property of the ERC-20 standard.

A-04 Payable constructor with no withdrawal path

■ INFORMATIONAL

Location. `StandardBEP20.sol:849`, the payable modifier on the constructor.

The constructor accepts native coin, but the contract has no function that can send it out. The template's `ServicePayer` contract, which would have forwarded such a payment, is declared in the file but not inherited. Any BNB sent at deployment would therefore be locked permanently. Slither reports this as `locked-ether`. The deployment transaction carried a value of 0 and the contract's native balance is 0 today, so no funds are affected.

Recommendation. None for this deployment. For future deployments from the same template, drop payable or inherit the payment forwarder deliberately.

Status. Acknowledged. No impact on the live contract.

Every statement in the original report was checked again on 22 September 2026. The findings stand. The following items were changed; the full claim-by-claim record is kept with this report.

06 Verification evidence

ITEM	VALUE READ FROM THE CHAIN
Contract	0x51b27860c3432a72e24a7e0a98916d10cb716d5c
Network	BNB Smart Chain, chainId 56, block 123,364,658
name / symbol / decimals	AlphaAdvisor / APAD / 18
totalSupply	10000000000000000000000000 base units, 1,000,000,000 APAD
Deployment	block 59,264,389, 29 August 2025 05:49 UTC, 0xaa8c3009...37f7933d
Deploying account	0xB4A6b7DF3e179421768459C10017e225699DB2f7
First movement of supply	block 80,907,460, 13 February 2026 03:40 UTC, 0x5fb72709...0ca88895, to 0xa804e25b8bf22161d9a415a59ae38be60442458a

Ownership transfer	block 81,141,126, 14 February 2026 08:53 UTC, 0x87938001...899f07f8
Current owner	0xa9e3e9AfA056D200241B85AEC824B039Ffe17582 , no contract code, 795,929,872 APAD (79.59% of supply)
Native coin held by the token	0 BNB
Runtime bytecode	4,277 bytes, 113 PUSH0 opcodes (prague target)
Verification	Sourcify: creationMatch = match, runtimeMatch = match (verified 18 March 2026)

Test card, executed 22 September 2026

GROUP	CASES	RESULT
Metadata, supply, zero-supply guard, transfer semantics and ERC-6093 errors	11	■ CONFIRMED
Allowance semantics, infinite allowance, approval overwrite	5	■ CONFIRMED
Owner role: initial owner, getOwner, restrictions, transfer, renounce, no power over balances	8	■ CONFIRMED
Absent selectors, native coin rejected after deploy, garbage calldata	10	■ CONFIRMED
Payable constructor locks value; no callback to recipients	2	■ CONFIRMED
Fuzz properties at 512 runs: supply constant, conservation, unauthorised transferFrom, ownership takeover, allowance fidelity	5	■ CONFIRMED
Total	41	41 passing, 0 failing

Static analysis, run 22 September 2026

DETECTOR	RESULT
locked-ether	Payable constructor without a withdrawal function. Finding A-04.
dead-code	_msgData, _contextSuffixLength and _burn are never used. Inherited library code; confirms there is no burn path.
solc-version	The ^0.8.20 pragma range includes compiler releases with known issues. The deployed build used 0.8.30, which is outside the affected range for the listed issues.

07 Conclusion

The APAD contract is a plain fixed-supply token built from audited OpenZeppelin components, and its deployed code matches its published source. No holder balance can be changed by anyone other than the holder, and no function can create or destroy tokens.

The rating stays LOW. What a holder should weigh is concentration, not code: one externally owned account holds the owner role and 795,929,872 APAD (79.59% of supply), and the contract enforces no vesting. We recommend moving the owner role and unreleased allocations to a multi-signature wallet and vesting contracts, and publishing those addresses.

08 Appendix: project-specific source

The project-specific parts of the verified source, with the line numbers of the verified file. Everything else comes from unmodified OpenZeppelin libraries.

```
contract StandardBEP20 is ERC20Detailed, BEP20TokenOwner {
    constructor(
        string memory name_,
        string memory symbol_,
        uint8 decimals_,
        uint256 initialBalance_
    )
        payable
        ERC20Detailed(name_, symbol_, decimals_)
    {
        require(initialBalance_ > 0, "Initial supply cannot be zero");

        _mint(_msgSender(), initialBalance_);          // line 854
    }
}

abstract contract BEP20TokenOwner is Ownable {
    constructor() Ownable(_msgSender()) {}            // line 747

    function getOwner() external view returns (address) {
        return owner();
    }
}
```

09 Disclaimer

This report was prepared by SOKEN LTD (Companies House 15253906). The audit was performed in December 2025; this edition re-verifies its statements as of 22 September 2026 at block 123,364,658. It covers the contract source and deployed bytecode at the address stated. It is a point-in-time technical review, not a guarantee that the code is free of defects, and it is not financial, investment or legal advice. On-chain values change, so re-read any figure at the linked address or transaction before relying on it. Off-chain conduct, key management, distribution decisions, listings and third-party integrations are outside the scope of this review.